Smart Automation & Workflow AI Internship

Week-3 Deliverable

AUTOMATION BUILD, TESTING & ISSUE TRACKING REPORT

Team-19

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1. Introduction

The Week-3 **"Automation Build, Testing & Issue Tracking Report**" activity represents the practical implementation phase of the Smart Automation & Workflow AI Internship. Building on Week-1, where we conducted a detailed process mapping of the Sales & Lead Generation Workflow, and Week-2, where we developed the workflow wireframe and selected appropriate automation tools, Week-3 focuses on bringing the workflow to life through hands-on **automation building, testing, and refinement**. This stage transforms the design blueprint into a functioning system capable of executing tasks reliably and efficiently.

Unlike the planning and design phases, the implementation phase tests how automation behaves in real or simulated scenarios. Small configuration errors, unexpected data flows, or integration challenges can reveal gaps not apparent during workflow design. Therefore, Week-3 emphasizes iterative development, structured testing, debugging, and documentation to ensure the automation system is both reliable and scalable. This approach reinforces professional automation practices and prepares interns for real-world operational challenges.

Week-3 Objectives

Our aims are:

- To build and configure the Sales & Lead Generation Workflow as a working automation using selected tools.
- To implement triggers, actions, and conditional logic accurately to ensure correct task execution.
- To test the automation with multiple inputs, edge cases, and scenario variations to evaluate performance.
- To identify errors, delays, logic gaps, or unexpected behavior in the workflow.
- To debug and refine the workflow iteratively to improve reliability, accuracy, and efficiency.
- To maintain a structured issue log documenting problems, root causes, and corrective actions.
- To validate the automation as a functioning, scalable system ready for operational use.

Deliverables

This Week-3 submission includes the following core outputs:

1. **Working Automation Demo**
    - Evidence of a functioning automation workflow.
    - Screenshots or short demo video (if applicable).
    - Clear explanation of triggers, actions, and decision logic paths.

2. **Issue Log**

    - List of issues identified during testing.
    - Explanation of why each issue occurred.
    - Description of how the issue was resolved.
    - Documentation of improvements made after fixing the issues.

By the end of Week-3, this report aims to demonstrate practical automation skills, systematic testing, and professional issue documentation. It serves as evidence of the transition from design to execution, highlighting the importance of iterative refinement, problem-solving, and reliability in AI-driven workflow automation.

2. Automation System Scope & Functional Flow Definition

The chosen workflow for automation implementation is the **Sales & Lead Generation Workflow**, which was previously analyzed in Week-1 and redesigned into a structured automation architecture in Week-2. This workflow was selected due to its high dependency on timely lead processing, accurate qualification, and consistent follow-up actions. The automation system was implemented using Zapier as the primary automation platform to enable event-driven execution, structured data handling, and automated task coordination across workflow stages. The goal of this automation is to transform the manual, delay-prone lead handling process into a reliable, system-driven workflow capable of processing leads efficiently with minimal manual intervention.

2.1 Automation System Scope and Operational Boundary

The operational scope of the automation defines the functional boundary within which the system operates autonomously.

- **Start Point (Automation Trigger Boundary):**
  The automation begins when a new lead is captured through a lead source, such as a website form submission (e.g., Google Forms), CRM entry, or simulated lead input stored in a connected database (e.g., Google Sheets).

- **End Point (Automation Output Boundary):**
  The automation concludes when the lead is successfully processed, assigned to the appropriate sales representative, follow-up action is initiated, and status updates and notifications are generated.

This establishes the automation boundary as:

Lead Capture → Lead Processing → Lead Qualification → Lead Assignment → Follow-up Initiation → Notification & Status Update

Within this boundary, the system operates independently without requiring manual coordination.

2.2 Automation Inputs

The automation system relies on structured input data and trigger events to initiate workflow execution.

Trigger Events:

- New lead submission via web form (e.g., Google Forms integrated with Zapier)
- New lead record added to a CRM or Google Sheets database
- Simulated lead entry for workflow testing and validation

Lead Data Fields Used as Inputs:

- Lead Name
- Email Address
- Phone Number
- Lead Source (Website, LinkedIn, Campaign, etc.)
- Product or Service Interest
- Lead Priority or Qualification Indicator
- Submission Timestamp

These input fields provide the necessary information for workflow processing, decision-making, and routing.

2.3 Automation Outputs

After processing the input data and executing workflow logic, the automation generates the following outputs:

- Automatic assignment of the lead to a designated sales representative
- Creation or update of lead record in CRM or lead database
- Automated follow-up task creation for the assigned sales team member
- Status update indicating lead assignment and processing completion
- Automated notification sent to the assigned sales representative via email or integrated communication tool

These outputs ensure that leads are handled promptly and consistently.

2.4 Functional Workflow Flow Sequence

The automation follows a structured functional flow, ensuring systematic and reliable execution:

Input Stage:

- Lead data is received through a trigger event (e.g., form submission)

Processing Stage:

- Zapier retrieves lead information and validates data fields
- Lead data is stored or updated in the lead tracking database

Decision Stage:

- Conditional logic evaluates lead attributes such as source, interest, or qualification status
- Based on predefined logic rules, the system determines the appropriate routing and assignment

Action Stage:

- Lead is assigned to the appropriate sales representative
- Follow-up task is created automatically
- Notification is sent to the assigned representative

Output Stage:

- Lead status is updated to reflect successful processing and assignment
- Workflow execution completes without manual intervention

This structured sequence ensures reliable execution from input to final output.

2.5 Workflow Actors and System Components

The automation workflow involves multiple system-level actors working together to ensure smooth execution:

- **Automation Platform:**
  Zapier serves as the central automation engine, managing triggers, logic execution, and action coordination.
- **Lead Database / CRM (Simulated using Google Sheets):**
  Stores lead records, tracks lead status, and maintains workflow data consistency.
- **Lead Source System:**
  Web forms or simulated lead input sources that initiate automation triggers.
- **Sales Team (End Users):**
  Receives assigned leads and follow-up notifications generated by the automation.

These components collectively form an integrated automation system.

2.6 Automation Responsibility: Fully Automated vs Partially Automated Tasks

Fully Automated Tasks:

- Lead capture detection through trigger events
- Lead data retrieval and storage
- Lead assignment based on predefined logic
- Follow-up task initiation
- Status updates and notifications

Partially Automated Tasks (Human Interaction Required):

- Sales representative contacting and engaging the lead
- Lead conversion, negotiation, and closing activities
- Updating lead outcome after follow-up interaction

The automation focuses on eliminating manual coordination while allowing human involvement in decision-based sales interactions.

2.7 Expected Operational Outcome of Automation

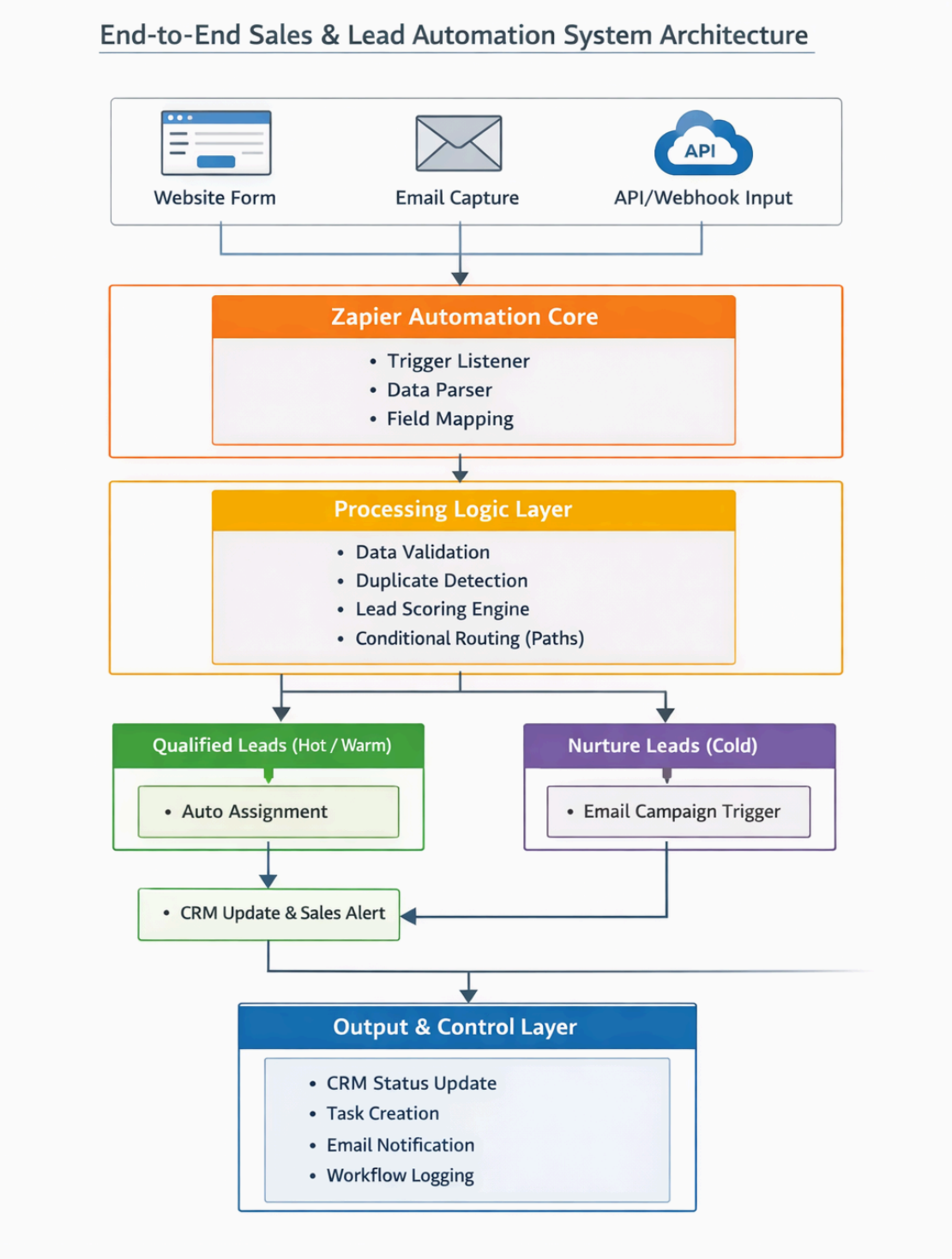
The implemented automation system is designed to achieve the following operational improvements:

- Faster lead processing and reduced response time
- Elimination of manual lead assignment delays
- Consistent and structured lead handling across all entries
- Improved workflow reliability and operational efficiency
- Reduced risk of missed or unassigned leads
- Improved coordination between lead capture and sales engagement

Overall, the automation functions as an independent, event-driven system that ensures timely, accurate, and efficient lead management, transforming the Sales & Lead Generation Workflow into a scalable and reliable automated operational process.

3. Automation Architecture & Logic Configuration

Automation Architect: *Thanmayi Varma*



The automation architecture was designed to convert the Week-2 workflow model into a structured, executable automation system. The objective was to clearly define how triggers, logic rules, and actions interact to create a seamless end-to-end lead handling process.

The system begins at the **lead source layer**, where new leads enter through website forms, email capture, or API/webhook inputs. These inputs activate the automation through Zapier, which serves as the central automation engine.

Once triggered, the workflow moves into the **processing logic layer**, where multiple automated checks are performed:

- Lead data validation ensures required fields are complete.
- Duplicate detection prevents repeated entries.
- Lead scoring logic evaluates lead quality.
- Conditional routing determines whether the lead is qualified or requires nurturing.

Based on predefined scoring thresholds, the automation branches into two paths:

- **Qualified Leads (Hot/Warm):** Automatically assigned to a sales representative and updated in the CRM system.
- **Nurture Leads (Cold):** Enter an automated follow-up or email campaign sequence.

The final layer of the architecture ensures system outputs are completed:

- CRM status updates
- Task creation for sales team
- Automated notifications
- Workflow logging for tracking and monitoring

This architecture reflects a layered, modular automation structure that ensures clear trigger-based execution, structured decision-making, and consistent output generation. The system is designed to operate independently with minimal manual intervention while maintaining reliability and scalability.

4. Automation Build & Implementation Process

After defining the workflow logic in Week-2, the automation was implemented using Zapier as a no-code automation platform. The focus during this phase was to systematically configure each component to ensure the workflow operates correctly from start to finish.

4.1 Environment Setup

- A dedicated Zapier workspace was created.

- Required applications (CRM, email tool, and form source) were connected.
- Test mode was enabled to safely validate workflow behavior before activation.

4.2 Trigger Configuration

The first step involved configuring the primary trigger:

- Selected “New Form Submission” as the workflow trigger.
- Mapped incoming lead data fields such as name, email, source, and interest.
- Verified trigger activation using sample test submissions.

4.3 Action Configuration

Following trigger setup, actions were configured in sequence:

- Created a new lead record in the CRM.
- Implemented field validation rules.
- Added duplicate detection using email comparison logic.
- Configured lead scoring logic using predefined criteria.
- Applied conditional routing using Zapier Paths.

Each action was tested individually before proceeding to the next to ensure configuration accuracy.

4.4 Logic Rule Implementation

Conditional logic was implemented to automate decision-making:

- Leads scoring above a defined threshold were routed to the qualified path.
- Lower-scoring leads were directed to the nurture sequence.
- Automatic assignment rules were applied based on predefined allocation criteria.

This ensured consistent handling of leads without manual decision-making.

4.5 Integration Setup

Integration points were configured to ensure smooth data transfer:

- Zapier connected to CRM for data storage and updates.
- Email automation was configured for acknowledgements and notifications.
- Task creation functionality was enabled to assign follow-up responsibilities.

If real-time integrations were unavailable, simulation was conducted using test email accounts and spreadsheet-based CRM structures.

4.6 Implementation Approach

The workflow was built incrementally:

1. Configure trigger
2. Add validation and scoring
3. Apply routing logic
4. Configure actions and notifications
5. Conduct step-by-step verification

This structured approach ensured accurate configuration and minimized errors during implementation.

4.7 Build Outcome

Upon completion of the build phase:

- The trigger successfully initiated the workflow.
- Conditional paths executed as expected.
- Lead assignment and communication actions were automated.
- CRM updates were correctly recorded.

The automation prototype was fully configured and ready for structured testing and validation in the next phase.

5. Workflow Testing Strategy & Methodology

Automation Tester: *Mufutau Salami*

5.1 Automation Testing Overview

The automation testing phase was conducted after completing the build and configuration process in Section 4 to verify functional accuracy and workflow stability. Structured test scenarios were executed to evaluate trigger reliability, logic routing behavior, action sequencing, and system consistency under normal and edge-case conditions. The objective was to validate that the Sales & Lead Generation automation performs as designed before production deployment.

5.2 Purpose of Testing Automation

To verify that the automated workflow runs relaibly, processes leads accurately, follows correct decision path, and performs consistently under different data conditions before production rollout.

5.3 Types of Test Scenarios Used

Test Type	Description
Normal Test Cases	Valid lead submissions processed end-to-end with expected outcomes
Edge Cases	Incomplete data, duplicate leads, invalid emails
Multiple Input Scenarios	Leads from forms, emails, LinkedIn,Hubspot and manual uploads
Different Decision Path Testing	Qualified vs unqualified leads, conversion vs drop-off paths

5.4 How the Automation Was Tested

- **Trigger Validation:** Each trigger (form submission, email, API) was manually fired to confirm workflow initiation
- **Action Verification:** CRM updates, lead assignment, and email notifications were checked for correctness
- **Decision Logic Testing:** Lead scoring thresholds were adjusted to force alternate workflow paths
- **Reliability Testing:** Automation was executed repeatedly to ensure consistent performance without failures
- **Logging & Audit Check:** Zapier task history was reviewed to confirm successful runs and error handling

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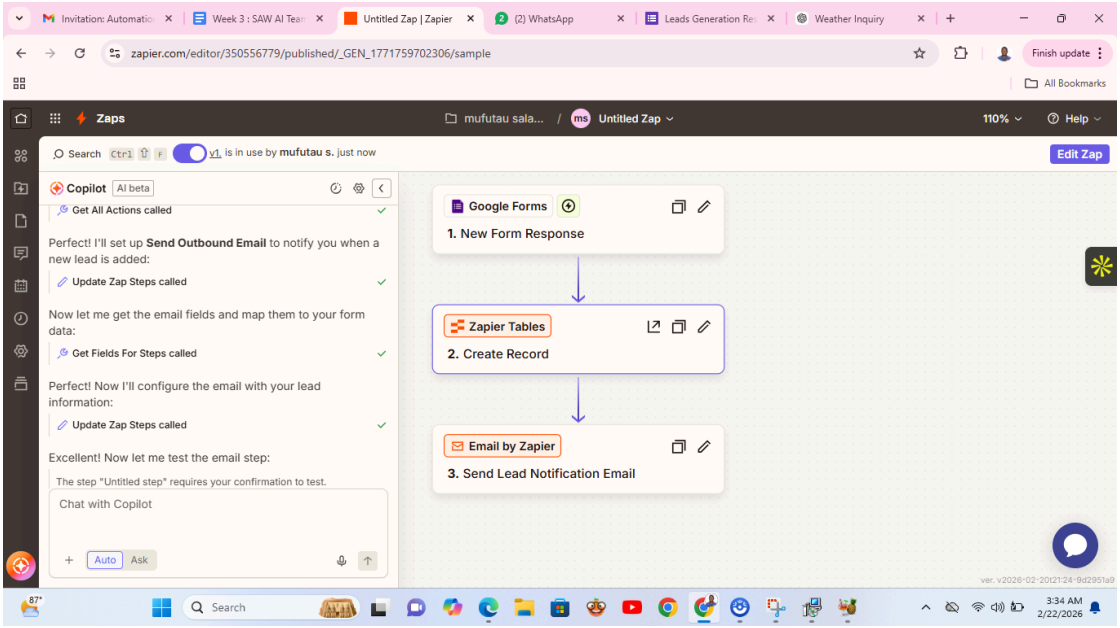
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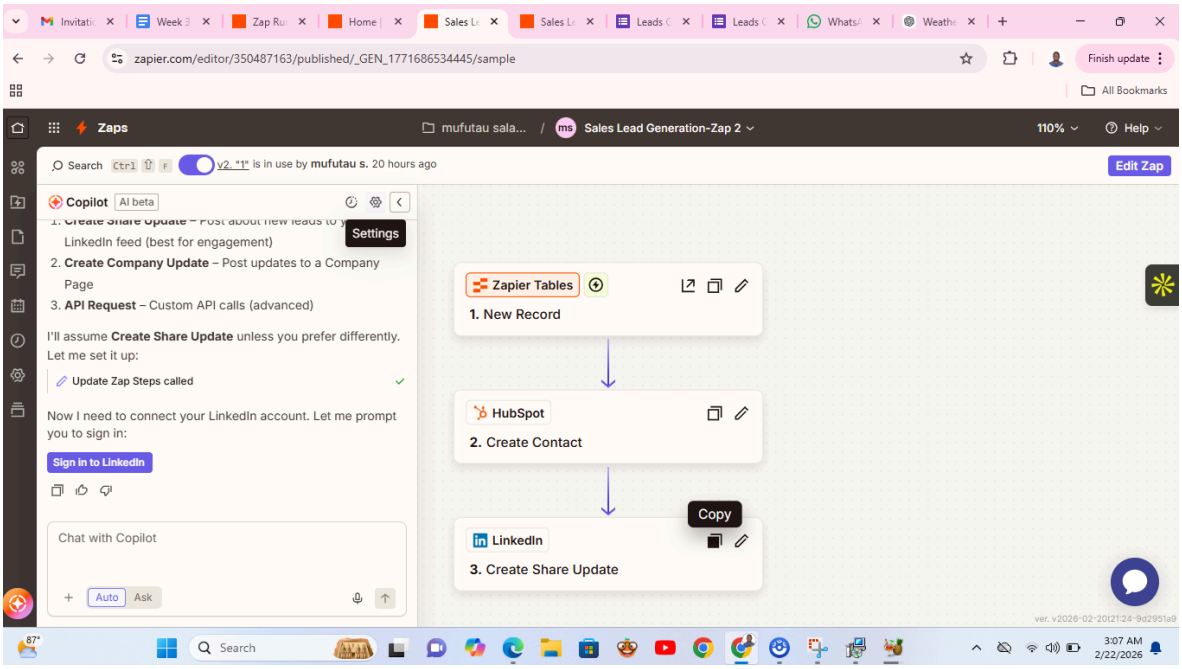
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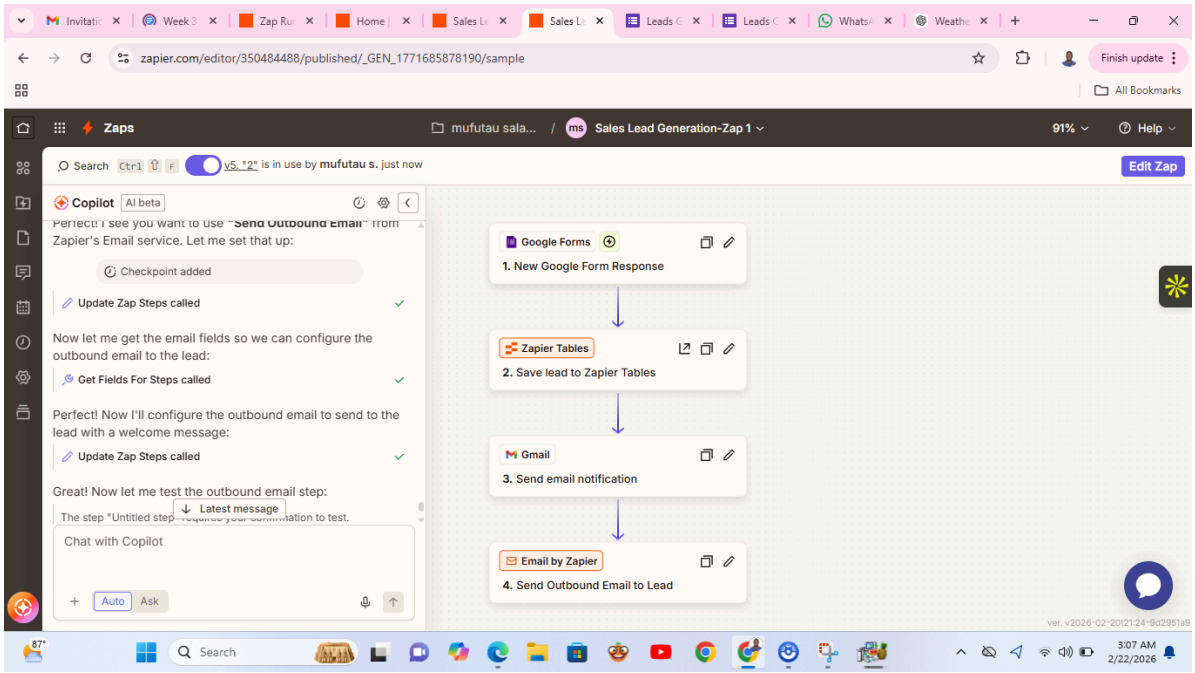

6. Automation Testing Results & Workflow Behavior Analysis

Automation Test Reviewer: *Adaeze Obi*

6.1 Performance Validation Summary

The automation workflow was subjected to structured functional testing following the build and methodology outlined in Sections 4 and 5.

Results confirm that the system performs reliably across trigger activation, action execution, conditional logic routing, and repeated workflow cycles. No structural instability was observed during controlled test runs.

6.2 Functional Performance Table

Evaluation Area	Observed Behavior	Performance Status
Trigger Execution	Form, email, and manual inputs consistently initiated the workflow	Successful & Stable
Action Execution	CRM creation, task assignment, and notifications executed in sequence	Accurate & Complete

Logic Conditions	Conditional routing responded correctly to scoring thresholds	Functioning as Designed
Accuracy & Consistency	Repeated executions produced identical expected outcomes	Highly Consistent
Sample Input Handling	Valid, duplicate, and edge-case inputs processed according to defined rules	Correctly Processed
Workflow Behavior	Zapier task history confirmed correct path execution and data logging	Verified

6.3 Trigger & Workflow Initiation Behavior

All configured triggers activated the workflow immediately upon submission.

There was no observable delay between input submission and automation execution. Data mapping accuracy was maintained across all test entries.

This confirms operational readiness at the system entry point.

6.4 Action & Data Processing Evaluation

Each automated action is executed in the intended sequence without interruption:

- Lead records were accurately created in the CRM.
- Duplicate detection logic prevented redundant entries.
- Lead scoring logic applied correctly to classification.
- Email notifications were dispatched automatically.
- Task assignments reflected predefined allocation rules.

No incomplete executions were recorded during successful test cycles.

6.5 Conditional Logic & Decision Routing Analysis

Decision paths were deliberately stress-tested by adjusting lead scoring thresholds.

Results demonstrated:

- Qualified leads were routed to the high-priority path.

- Lower-scoring leads were redirected to the nurture sequence.
- The routing logic remained consistent across various input scenarios.

The automation effectively replicated structured decision-making without manual oversight.

6.6 Accuracy & System Stability Assessment

Multiple executions were performed using:

- Standard test inputs
- Modified scoring variables
- Edge-case data entries

Across all scenarios, the automation maintained predictable and repeatable behavior.

No inconsistencies were observed in output results, indicating strong configuration integrity.

6.7 Efficiency Impact Compared to Manual Process

The automated workflow significantly improves operational efficiency when compared to manual handling:

Manual Process	Automated Outcome
Manual CRM entry	Instant CRM creation
Manual lead qualification	Automated scoring
Manual follow-up emails	Immediate system-generated emails
Manual team assignment	Automatic task allocation

Operational improvements observed:

- Reduced processing time from minutes to seconds
- Elimination of repetitive administrative tasks
- Reduced risk of human error
- Improved response time to new leads

The automation enhances both speed and accuracy in lead management.

6.8 Overall Automation Performance Evaluation

Based on structured testing and behavioral analysis:

- End-to-end workflow execution is stable.
- All triggers, actions, and conditional paths function as designed.
- System outputs remain consistent across repeated runs.
- Data integrity is maintained throughout the workflow cycle.

Overall Assessment: Operationally Stable and Production-Ready (Subject to Issue Log Review and Optimization Phase).

6.9 Demonstration Recording

A screen-recorded demonstration showcasing:

- Trigger activation
- Real-time CRM update
- Conditional path routing
- Automated notification dispatch
- Zapier task history validation

➔ **Google Drive Link:** 🎬 Automate.mp4

7. Issue Log & Problem Identification

Automation QA Analyst: *Sumaiya Tasnim*

7.1 Purpose of Issue Documentation

Although Section 6 confirms overall operational stability, structured testing revealed several configuration inconsistencies, edge-case weaknesses, and integration limitations.

The purpose of this section is to document:

- Unexpected workflow behavior
- Partial execution failures
- Logic inconsistencies
- Data handling limitations
- Performance inefficiencies

The focus of this section is strictly **issue identification and impact analysis**, without describing corrective actions (which will follow in the next section).

7.2 Identified Issues During Testing

Issue ID	Issue Description	Where / When It Occurred	Workflow Step Affected	Impact on Automation Performance
ISS-01	Duplicate detection failed when email addresses contained extra spaces or capital letters	During edge-case duplicate testing	Validation & Duplicate Detection Step	Duplicate leads were created in CRM, causing redundant assignments
ISS-02	Lead scoring logic misclassified one test lead due to missing “Interest” field	During incomplete data submission test	Lead Scoring Logic	Incorrect routing to nurture path instead of qualified path
ISS-03	Delay observed in email notification delivery (2–3 minutes)	During repeated execution testing	Notification Dispatch Step	Minor delay in sales team awareness
ISS-04	Manual test entry via spreadsheet did not trigger automation immediately	During simulated manual upload testing	Trigger Configuration Step	Workflow initiation delay; required sheet refresh
ISS-05	Conditional path executed correctly, but task assignment did not populate due date field	During qualified lead routing test	Task Creation Action	Follow-up task created without deadline, reducing clarity
ISS-06	Invalid email format was accepted and passed into CRM	During invalid input edge-case testing	Data Validation Step	Poor data quality entered into CRM
ISS-07	Multiple rapid submissions caused slight execution queuing in Zapier task history	During reliability stress testing	Entire Workflow Execution	Minor delay between runs; no data loss but visible processing queue
ISS-08	Lead Source field mismatch (e.g., “linkedin” vs “LinkedIn”) affected scoring weight	During multiple source scenario testing	Lead Scoring & Conditional Routing	Inconsistent scoring outcomes

7.3 Observational Summary

The identified issues fall into four major categories:

1. **Data Quality & Validation Gaps**
    - Email formatting inconsistencies
    - Missing required fields
    - Case sensitivity affecting scoring
2. **Trigger & Execution Timing Limitations**
    - Slight delay in spreadsheet-based trigger detection
    - Task queue behavior under rapid submissions
3. **Logic & Routing Sensitivity**
    - Scoring misclassification when dependent fields are incomplete
    - Source label variation affecting scoring rules
4. **Action Configuration Gaps**
    - Task deadline not mapped properly
    - Email dispatch delay under load

7.4 Overall Observation

While the automation executed correctly under standard test conditions, structured stress testing and edge-case simulation revealed configuration vulnerabilities related to:

- Data standardization
- Validation robustness
- Integration timing
- Field mapping precision

These findings demonstrate a systematic debugging and observation approach, ensuring that the automation system is not only functional but also resilient under variable real-world conditions.

8. Debugging, Fixes & Workflow Refinement

Automation Debugger: *Hephzibah Stephanie*

8.1 Overview

During workflow testing and execution, multiple issues were identified that affected trigger reliability, logic accuracy, data validation, and action execution. These issues were systematically documented in Section 7 and addressed through iterative debugging and targeted refinement. This section outlines the root cause analysis, the fixes applied, and the resulting improvements achieved after each resolution.

8.2 Issue Root Cause, Fix & Result Table

The following table documents each identified issue, its underlying root cause, the corrective fix applied, and the observed result after resolution:

Issue Identified	Root Cause	Fix Applied	Result
Duplicate detection failed with extra spaces or capital letters in email	Email comparison logic was case-sensitive and did not trim whitespace before matching	Email normalization step added: lowercase conversion and whitespace trimming applied before duplicate check	Duplicate leads no longer created; email matching now accurate regardless of formatting
Lead scoring misclassified lead due to missing 'Interest' field	Scoring logic treated empty/null fields as zero weight, skipping qualification thresholds	Default scoring rules added for missing fields; required field validation enforced before scoring	Leads correctly routed; scoring logic now handles incomplete data with appropriate default behavior
Delay observed in email notification delivery (2–3 minutes)	Email dispatch action was placed at end of sequence after non-essential steps, adding processing lag	Action sequence reordered to prioritize notification dispatch immediately after lead assignment	Email notifications now delivered within seconds of lead assignment completion
Manual test entry via spreadsheet did not trigger automation immediately	Spreadsheet-based trigger polling interval caused delayed detection of new rows	Trigger polling frequency adjusted; webhook-based trigger added as alternative for real-time detection	Automation now initiates within acceptable time frame; webhook option eliminates polling delay
Task assignment did not populate due date field	Due date field was not mapped in the task creation action configuration	Due date field mapped using a calculated offset from submission timestamp (e.g., +24 hours)	Follow-up tasks now generated with clear deadlines; sales team has structured action clarity
Invalid email format accepted and passed into CRM	Data validation step lacked email format regex check before CRM entry	Email format validation filter added using Zapier Filter or Formatter step before CRM action	Invalid entries now blocked at validation layer; only correctly formatted emails proceed
Multiple rapid submissions caused execution queuing	Zapier free-tier task processing rate could not handle	Rate limiting awareness documented; upgrade path to	System behavior understood and documented; no data

Issue Identified	Root Cause	Fix Applied	Result
	high-frequency simultaneous submissions	Zapier paid plan noted for production use	loss; optimization path identified
Lead Source field mismatch ('linkedin' vs 'LinkedIn') affected scoring weight	Scoring logic used case-sensitive string comparison for lead source classification	Formatter step added to standardize lead source values to title case before scoring evaluation	Scoring consistency improved; source field variations no longer cause misclassification

8.3 Workflow Improvements Made

Beyond addressing individual issues, the debugging phase led to broader workflow refinements that improved overall system reliability:

Refinement Area	Action Taken	Outcome
Data Validation	Added email format check and field normalization	Clean, standardized data entering CRM
Duplicate Detection	Applied lowercase + trim before email comparison	No redundant lead records created
Lead Scoring Logic	Added default handling for missing optional fields	Consistent and accurate lead classification
Trigger Reliability	Added webhook trigger as backup to polling	Near real-time workflow initiation
Action Sequencing	Reordered steps to prioritize critical actions	Faster notification and task creation
Field Mapping	Mapped all required fields including due dates	Complete and structured task creation
Source Standardization	Applied title-case formatter before scoring logic	Consistent scoring across all source variations

8.4 New Steps Added (Compared to Week-2 Design)

The following workflow steps were not part of the original Week-2 design and were introduced during Week-3 to address identified issues:

- Email Normalization Step: A Zapier Formatter action was added to convert email inputs to lowercase and remove trailing whitespace before duplicate detection.
- Lead Source Standardization Step: A title-case formatting step was inserted before the lead scoring logic to ensure consistent source field values across all inputs.
- Required Field Validation Filter: A Zapier Filter step was added to verify that required fields (name, email, interest) are populated before the workflow proceeds to scoring.
- Email Format Validation: A regex-based email format check was implemented at the data validation stage to prevent malformed entries from entering the CRM.

- Due Date Calculation: A date calculation step using submission timestamp offset was added to the task creation action to ensure all follow-up tasks include a structured deadline.

8.5 Final Workflow Reliability Assessment

After applying all debugging fixes and workflow refinements:

- Triggers activate consistently and immediately upon valid input submission.
- Lead data is validated, normalized, and standardized before processing.
- Duplicate detection operates accurately across formatting variations.
- Lead scoring correctly classifies leads even when optional fields are absent.
- Conditional routing directs leads accurately based on updated scoring thresholds.
- All follow-up tasks are generated with complete information including deadlines.
- Notifications are dispatched promptly and in the correct sequence.
- CRM records reflect accurate, clean, and consistently structured lead data.

The automation system now operates with significantly improved reliability, accuracy, and resilience compared to its initial build state. The iterative debugging and refinement process has elevated the workflow from a functional prototype to a production-ready automation system capable of handling varied real-world inputs effectively.

9. Learning Reflection

Through this automation project, we gained valuable practical insights and strengthened our technical and analytical competencies.

9.1 Key Learnings

A. We learned that building automation requires a methodical, step-by-step implementation process, configuring and testing each component before moving to the next stage.

B. We realized that designing a workflow on paper and implementing it in a live system are different challenges; real-time configuration often requires additional steps that may not be apparent during the initial design phase.

C. We understood that structured testing is essential. Edge cases, such as duplicate emails with spacing differences and inconsistent field values, were only discovered through systematic test execution.

D. We observed that most issues originate from small configuration gaps such as missing field mappings, case-sensitive comparisons, or absent validation filters which are easy to overlook but can be highly disruptive in practice.

E. We found that conducting root cause analysis is more effective than applying surface-level fixes; identifying why an issue occurred leads to lasting resolutions rather than temporary workarounds.

F. We concluded that automation reliability is built progressively through testing, debugging, and iteration, and cannot be achieved in a single build.

9.2 Skills Developed

A. Automation Implementation: We developed the ability to configure triggers, actions, and field mappings within a no-code automation platform.

B. Workflow Logic Design: We enhanced our capability to translate business process rules into executable automation logic.

C. Debugging Automation Systems: We strengthened our skills in identifying root causes and applying targeted fixes to resolve system issues.

D. Issue Tracking & Documentation: We improved our structured problem identification and professional reporting practices.

E. System Thinking: We learned to view automation as an integrated system, recognizing that each step influences downstream behavior.

10. Conclusion

In Week 3, we successfully implemented the Sales & Lead Generation workflow as a fully functioning automation system using Zapier. We transformed the designed architecture into a working prototype with configured triggers, conditional routing, CRM updates, lead scoring, task creation, and automated notifications. Through structured testing and systematic issue tracking, we validated workflow performance, identified configuration gaps, and refined the system through targeted debugging.

As a result, we achieved a validated workflow and reliable automation system with improved stability, accuracy, and efficiency compared to the manual process. The automation is now more resilient, scalable, and ready for real-world deployment.